Research Square

Preprints are preliminary reports that have not undergone peer review.
They should not be considered conclusive, used to inform clinical practice,
or referenced by the media as validated information.

Rapid Assessment of Olive Oil Adulteration Using LIF Spectroscopy and a Comparative Study of Machine Learning Models.

le li
Shandong Jiaotong University
Lanjun Sun
sun.lanjun@163.com

Shandong Jiaotong University
Xiongfei Meng
Shandong Jiaotong University
Zhijian Liu
Shandong Jiaotong University
Shuhan Huang
Shandong Jiaotong University

Research Article

Keywords: Olive oil adulteration, Laser-induced fluorescence, AlexNet, Deep Learning

Posted Date: December 4th, 2025

DOI: https://doi.org/10.21203/rs.3.rs-8152295/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License.
Read Full License

Additional Declarations: No competing interests reported.

Abstract

Laser-induced fluorescence (LIF) provides a rapid, non-destructive tool for detecting adulteration in olive oil. However, severe spectral overlap remains a major obstacle, hindering both qualitative identification and quantitative determination of adulteration levels. In this study, a 405 nm diode laser source was used to excite pure extra virgin olive oil (EVOO), soybean oil, peanut oil, and corn oil, as well as binary blends of these three vegetable oils in EVOO, and a total of 1,140 sets of fluorescence spectral data were obtained. Convolutional neural network (CNN), long short-term memory network (LSTM), and improved deep convolutional neural network (AlexNet) were respectively employed for the detection and quantitative analysis of adulteration in olive oil. The models achieved 100% classification accuracy, robustly differentiating pure EVOO from adulterated oil, which confirms their complete reliability for detecting olive oil adulteration. In terms of quantitative prediction, AlexNet performs better than LSTM and CNN, with the coefficient of determination (R^2) of 0.9930 and root-mean-square error (RMSE) of 0.1258%. Combining LIF technology with the AlexNet deep learning model enables rapid detection of olive oil adulteration while allowing for precise quantification of adulteration levels, thereby offering an innovative approach to food quality monitoring.

1. Introduction

Rich in monounsaturated fatty acids and polyphenolic antioxidants, olive oil markedly reduces cardiovascular risk and exerts potent anti-inflammatory and antioxidant effects. In the global vegetable oil market, it has emerged as a cooking oil that combines nutritional functionality with economic value (Rahman et al., 2024). In particular, due to its strict production standards and distinctive flavor, EVOO commands a significantly more expensive market price than ordinary vegetable oil, making it a significant target for adulteration (Izquierdo et al., 2020). Approximately 30% of commercially available olive oils are adulterated to varying degrees. Common practices include blending in low-cost vegetable oils such as peanut oil (PO), soybean oil (SO), and corn oil (CO) to obtain illegal profits (Hashempour-baltork et al., 2024). Adulteration has been proven to undermine consumers' rights and interests, disrupt market order, and cause health risks due to oxidation byproducts from inferior oils (Alharbi et al., 2024). Consequently, developing a rapid, accurate, and user-friendly adulteration detection method is imperative to ensure the smooth and sustainable development of the olive oil industry and maintain a healthy market (Torreblanca-Zanca et al., 2019).

In recent years, olive oil adulteration detection technology has faced multiple challenges (Hooshyari & Casale, 2021). Researchers have proposed different methods for identifying vegetable oils, including gas chromatography-mass spectrometry (GC-MS) (Saitta et al., 2002), acid value and peroxide value testing (Yang et al., 2024), and high-performance liquid chromatography (HPLC) (Bongiorno et al., 2023). Despite their sensitivity and efficiency, these detection techniques face practical limitations such as complex sample preparation, high cost, and dependence on specialized knowledge, all of which render them suboptimal for rapid screening. Although the continuous development of electronic nose technology for detecting olive oil adulteration, these systems exhibit limited ability to distinguish

between similar volatile compounds (Moraes et al., 2025). This limitation becomes particularly pronounced when adulterants share olfactory characteristics with olive oil, which potentially leads to misclassification of different adulteration types. Spectroscopic techniques such as near-infrared (NIR) spectroscopy (Mendes et al., 2015), Fourier transform infrared (FTIR) spectroscopy (Uncu et al., 2019), and Raman spectroscopy offer advantages including non-contact operation (Arendse et al., 2021), rapid response, and non-destructive analysis, which have long been utilized for assessing the quality of vegetable oil. While widely used, NIR and FTIR spectroscopy rely on equipment that is often large and complex, with resolution levels that may not be ideally suited for detecting adulteration at low concentrations (< 5%). Raman spectroscopy offers high resolution but is susceptible to background signal interference in complex multi-component adulteration environments, which may result in reduced sensitivity. Consequently, there is an increasing demand for analytical techniques that are rapid, economical, and portable, with reduced reliance on sample preparation.

LIF is a highly sensitive, non-invasive technique that uses laser excitation to induce fluorescence in materials, allowing their properties to be determined from the emitted fluorescent signals (Gandhi et al., 2011). In marine environmental monitoring, the characteristic fluorescence spectra of pollutants like petroleum hydrocarbons are utilized by LIF to rapidly identify pollutant types, allowing for real-time monitoring of marine oil spills (Loh et al., 2021). In real-time urban air quality monitoring, LIF combined with lidar (LIF-LIDAR) enables remote detection of atmospheric pollutants such as NO_2, SO_2, and VOCs (Simard et al., 2004). LIF technology can also non-destructively excite pollutants present in soil, analyzing their distribution gradients while preserving soil structure integrity and quantifying pesticide residues in farmland (Tadini et al., 2021). When a specific wavelength laser excites vegetable oils, the natural fluorescent substances (such as vitamin E, chlorophyll derivatives, and polyphenolic compounds) produce characteristic emission spectra (Liang et al., 2025). Furthermore, the advent of deep learning algorithms in recent years has yielded novel solutions for the detection of LIF. The algorithm's advanced feature extraction and pattern recognition capabilities have effectively addressed the challenge of spectral peak overlap in spectral data, which markedly enhances the efficiency of data analysis (Li et al., 2023). For example, Sun et al. estimated surface features of marine oil spills using iterative adaptive density-based clustering (IA-DBSCAN) based on LIF technology (Sun et al., 2024). Yan et al. combined LIF with an extreme learning machine (ELM) optimized by the osprey optimization algorithm (OOA) classifier for qualitative identification of eight soil microplastic types, and used sequential projection algorithm (SPA) with partial least squares algorithm (PLS) regression for quantitative detection of PA66(Yan et al., 2024). Bavali et al. employed LIF spectroscopy combined with linear discriminant analysis (LDA), support vector machines (SVM), multilayer perceptron (MLP), and 1D-CNN learning algorithms to achieve quantitative analysis of sunflower oil adulteration in olive oil through multi-angle (20°-90°) spectral data collection (Bavali et al., 2025).

The integration of spectral technology and deep learning algorithms has substantially improved the analytical efficiency and classification accuracy of complex spectral data. CNN has achieved remarkable progress in image recognition through its powerful feature-extraction capabilities and has gradually been extended to spectral data. CNN can automatically learn from large-scale data and extract deep feature

information, circumventing the limitations of manual feature selection in traditional methods and significantly enhancing the model's generalizability and robustness (Yang et al., 2022). In addition to CNN, LSTM is a distinct type of recurrent neural network (RNN) that incorporates a distinctive gating mechanism to capture long-term dependencies in time series data effectively. The method has been demonstrated to be particularly well-suited for the analysis of temporal features in spectral data (Zhu et al., 2025). Researchers have proposed a series of derivative CNN models based on different training strategies and organizational structures. A classic example is AlexNet, which was proposed by Krizhevsky et al. in 2012(Krizhevsky et al., 2017). The rectified linear unit (ReLU) activation function was adopted to accelerate convergence; dropout regularization and data augmentation were employed to mitigate overfitting. Overlapping max-pooling was also used to enrich feature extraction. Hardware limitations were overcome, and training efficiency was improved through dual-GPU parallel computing. Subsequent studies have made modest structural adjustments and adapted AlexNet to spectra, which further advanced research in spectral data analysis. The introduction of these algorithms provides new approaches for processing complex spectral data and lays a solid foundation for achieving high-precision, automated identification and quantitative analysis.

In this study, LIF was utilized to acquire fluorescence spectra from olive oil samples exhibiting varying degrees of adulteration. Visualization and clustering of the vegetable oil samples in a three-dimensional space were performed using the t-SNE technique. Three classification models incorporating chlorophyll-based features were developed and evaluated, leading to accurate detection of adulterated olive oil. Furthermore, three quantitative models were established to predict the authenticity level of EVOO.

2. Materials and methods

2.1. Experimental setup

A LIF detection system was employed for the rapid detection of olive oil adulteration, and the experimental setup is illustrated in Fig. 1. The system is composed of three primary components: an emission system, a spectroscopic system, and a receiving system. The emission system uses a 405 nm diode laser as the excitation light source, with a peak output power of 2 W, and an adjustable repetition rate ranging from 100 to 10 kHz. The spectroscopic system employs the Ocean Optics HR-2000pro from Ocean Optics, in the United States, with a spectral detection range of 200 to 1100 nm and an optical resolution of 1 nm. The laser is incident on the oil sample at a 30° angle relative to the fiber probe. The oil sample absorbs the laser and generates a fluorescence signal. Fluorescent signal is filtered through an optical filter (optical density 0.001, transmittance 1.2%) to remove scattered laser light, then focused by a lens onto a fiber optic probe. The fluorescent signal is transmitted through the fiber optic to a spectrometer, which converts it into an electrical signal for transmission to a computer for analysis and processing.

2.2. Sample preparation

Oil samples used in this experiment included EVOO, PO, CO, and SO. The EVOO was Spain Borges brand, while the other three vegetable oils were Chinese Lu Hua brand. EVOO was blended with PO, CO, or SO to form binary blends in which the adulterant proportions ranged from 0% to 50% (0%, 5%, 10%, 15%, 20%, 30%, 50%, w/w). For each blend, three samples were prepared. The selected adulteration ratios (5%–20%) reflect the low-to-moderate adulteration levels most commonly encountered in commercial olive oil(Stavrakakis et al., 2022). The 30–50% adulteration levels simulated here are atypical in practice and serve primarily to probe the model performance limits(Temiz et al., 2021). During preparation, each binary blend was prepared in test tubes that had been rinsed with deionized water and dried. After shaking or ultrasonic treatment, the blend was transferred to a glass container, and its fluorescence intensity was measured and recorded, as shown in Fig. 2. To ensure data reliability, each condition was measured 60 times, yielding 1,140 fluorescence spectra in total. To prevent ambient light interference, measurements were carried out in a dark room at 23 ± 1℃(Chen et al., 2024).

2.3. Data preprocessing

Common interference factors during spectral data acquisition include system errors in the experimental setup and background noise (Malavi et al., 2024). These factors can cause background interference, baseline drift, inherent noise, and random noise, which can affect subsequent data analysis (Song et al., 2020). To eliminate excitation-wavelength interference, only the 412–800 nm segment (856 data points) was retained. Baseline correction was applied to the 1,140 spectra to remove background interference, which achieved preliminary purification of the spectral signal. Even after baseline correction, intrinsic noise and random noise from the spectrometer may still be present in the spectral signal. To further improve the quality of the spectral data, low-pass filtering was applied to suppress high-frequency noise and smooth the spectra. Savitsky-Golay smoothing was subsequently applied to suppress residual noise and accentuate the primary spectral features. The filtered spectra were normalized to the 0–1 intensity range. This step ensured that the spectral intensities of all samples were on the same rank, eliminating intensity variations caused by factors such as differences in instrument response.

2.4. Training and testing data

When conducting qualitative analysis of adulterated EVOO, three sample sets were constructed. These sets were designed to ensure a balanced dataset and evaluate the classifier's ability to identify EVOO with low, medium, and high levels of adulteration. Sample Group I included 60 pure EVOO and 180 adulterated EVOO (adulteration > 0%); Sample Group II included 60 pure EVOO and 540 adulterated EVOO (adulteration > 5%); Sample Group III included 60 pure EVOO and 360 adulterated EVOO (adulteration > 20%). In the quantitative analysis of olive oil adulteration levels, a total of 1080 samples (3×6×60) were included covering 3 categories of adulterated olive oil samples, with each category comprising 6 concentration levels and 60 replicate samples per concentration level. The sample set design is shown in Table 1.75% of the samples were selected at equal intervals according to the degree of adulteration as the training set, and the remaining 25% as the test set. The sample set was split through MATLAB’s cross-validation routine, and a hold-out validation was performed to ensure the model's robustness across various data subsets.

Table 1
Sample sets grouping design

Task	Group	Contains samples	Pure (0%)	Adulterated (5%-50%)	Level
Qualitative	I	0% 5%	60	3×1×60	[a]LA
	II	0% 10% 15% 20%	60	3×3×60	[b]MA
	III	0% 30% 50%	60	3×2×60	[c]HA
Quantitative	□	5%10%15%20%30% 50%	--	3×6×60	--

[a]LA: low adulteration [b]MA: moderate adulteration [c]HA: high adulteration

2.5. Algorithms and calculation steps

2.5.1. CNN

CNN is a deep learning model that draws inspiration from the biological visual system (Fan et al., 2019). The approach hinges on convolutional extraction of local features and a deep hierarchy that automatically learns multi-level representations. A typical CNN is composed of input layers, convolution (Conv) layers, pooling layers, activation function layers, fully connected (FC) layers, and output layers(Venturini et al., 2023). When the input dimension is reduced to a one-dimensional sequence, the traditional two-dimensional convolution kernel is replaced by a one-dimensional kernel that slides along a single axis, forming a 1D-CNN. The present study utilizes a 1D-CNN, the structural configuration of which is illustrated in Fig. 3(a). 1D-CNN takes the digitized data of the fluorescence spectrum as sequence input. The one-dimensional convolution kernel in the Conv layers slides along the spectral data dimension to extract local features such as spectral peak positions and intensities. Pooling layers are responsible for sampling the convolved features and retaining key information. Batch normalization (BN) layers normalize data within each batch, which adjusts the mean and variance of the data to a fixed range. The introduction of the ReLU activation function endows the network with nonlinear expressive capabilities, enabling it to capture complex pattern relationships in spectral data. Finally, the FC layers project the processed features into the output space, yielding the fluorescence spectrum analysis results and enabling effective processing of spectral numerical data.

2.5.2. LSTM

LSTM is a refined version of RNN, engineered to process sequential data with long-term dependencies. The traditional RNN suffers from vanishing or exploding gradients during training, which makes it difficult to effectively capture long-range dependency features in sequences (Li et al., 2023). By introducing input, forget, and output gates as gating mechanisms, LSTM can dynamically control the retention and transmission of information, which significantly enhances its ability to learn sequence structures. In this study, the LSTM model consists of three LSTM layers and ReLU activation layers, as illustrated in

Fig. 3(b). The model uses LSTM units to extract temporal features such as intensity changes at different wavenumber positions, reducing the data dimension while preserving critical temporal information. Subsequently, the ReLU activation function performs a nonlinear transformation on the extracted features, enhancing the model's ability to fit complex adulteration patterns. Finally, feature maps are projected onto the output space through FC layers to perform classification or regression analysis on spectral data.

2.5.3. AlexNet

AlexNet is a groundbreaking deep convolutional neural network in the field of deep learning. The model consists of five Conv layers, three max pooling layers, and FC layers, and is widely used in image recognition tasks (Lin et al., 2023). To enhance its applicability in spectral data processing, the classic AlexNet structure was optimized and improved, as shown in Fig. 3(c). We introduce a BN layer after the first Conv layers to enhance model stability and accelerate convergence. In the Conv layers design, to accommodate the sequential nature of spectral data and capture local features, a one-dimensional convolutional layer (Conv1D) is employed to replace the original two-dimensional convolution. Pooling layers reduce feature dimensions through maximum pooling operations, reducing computational load while preserving critical features. FC layers employ a dropout mechanism to randomly discard neurons, preventing overfitting. The single-output topology was replaced with a multi-output architecture that simultaneously classifies the blend and quantifies each constituent. This redesign fully exploits the inherent parallel-processing capacity of deep neural networks.

2.6. Model performance evaluation

All models were evaluated with 10-fold cross-validation; accuracy, precision, sensitivity, and specificity were computed from the confusion matrix and are shown in (1)–(4). The performance of the classifiers was evaluated by obtaining the receiver operating characteristic (ROC) curves. The area under the curve (AUC) was estimated, with higher AUC values indicating better classification performance (Leng et al., 2025).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Sensitivity = \frac{TP}{TP + FN} \quad (3)$$

$$Specificity = \frac{TN}{TN + FP}$$

4

TP denotes true positives that are correctly classified; FP denotes false positives that are incorrectly classified as positive; TN denotes true negatives that are correctly classified; FN denotes false negatives that are incorrectly classified as negative.

The performance of the adulteration regression models is demonstrated by box plots, which show the consistency between predicted and actual values. Model performance is quantitatively evaluated by the R^2 and RMSE, calculated in (5) –(6).

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

5

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

6

An R^2value closer to 1 indicates that the model explains a greater proportion of the variance in the target variable, reflecting a better fit to the data. The smaller the RMSE (> 0), the closer the predictions are to the true values and the higher the model's accuracy (Liu et al., 2023).

3. Results and discussion

3.1. Fluorescence spectra of oil samples

The spectrum of vegetable oil when excited at 405 nm reflects its inherent fluorescence characteristics. Spectra of same type oils exhibit similarities, but significant differences exist between different types. The 20% adulteration ratio exhibits the most pronounced difference between adulterated blends and pure vegetable oil, making it the preferred representative for adulteration. This strategy provides the most discriminative reference samples for subsequent visual analysis of binary blends. Figure 4 clearly shows the fluorescence signal differences between four pure vegetable oils and three binary blends with a 20% adulteration rate. EVOO exhibits a strong peak at 650–700 nm, attributable to chlorophyll groups. CO, SO, and PO show intense peaks at 450–600 nm, originating from the fluorescence of carotenoids, vitamin E, and fatty acid oxides commonly found in vegetable oils (Rohman & Man, 2011). The spectra of the three blends with 20% adulteration displayed only minor differences that are difficult to identify visually. Comparison of normalized spectra across all adulteration levels (Fig. 5) reveals that all spectra exhibit two primary bands: a broad, low-intensity band centered near 512 nm that increases with rising

adulterant concentration (carotenoids, vitamin E, and fatty acid oxidation products), while a narrow, high-intensity chlorophyll peak near 670 nm whose intensity declines as adulteration increases (Zhang et al., 2019).

3.2. Visualization and analysis of spectral data

The t-SNE visualization analysis was executed on the measured fluorescence spectra after smoothing and normalization to address peak overlap and spectral similarity. T-SNE is a frequently utilized nonlinear dimensionality reduction technique for spectral data. It projects high-dimensional data into low-dimensional space to reveal clustering and distributional structure. Figure 6 shows the clustering results of the t-SNE algorithm in a three-dimensional space, reproducing different clustering patterns corresponding to various types of oil products. The separation between all oil product clusters achieves significant inter-class differences while minimizing intra-class variation. Despite the strong spectral resemblance between CO and PO, as well as pure EVOO and its adulterated blend, t-SNE can still clearly separate each group. Minor fluorescence variations may affect the model's discrimination capability, making visual analysis of the spectrum necessary.

3.3. Classification Model Performance Evaluation

This study utilized CNN, LSTM, and AlexNet algorithms to identify pure EVOO and its adulterated binary blends. The three classifiers demonstrated remarkable efficacy in differentiating pure EVOO from adulterated EVOO and identifying the specific vegetable oil used for adulteration. Figure 7(a-c) shows the confusion matrices for the three models in distinguishing pure EVOO from adulterated EVOO. In these matrixes, 0 indicates pure EVOO, while 1 signifies adulterated EVOO. Among the 1,140 samples containing pure EVOO, pure EVOO was correctly classified in both the training and test sets with an accuracy of 100%, and no adulterated sample was misassigned as pure. For high-adulteration EVOO, LSTM and AlexNet achieved excellent classification with extremely low error rates and only a few samples misclassified, as shown in Fig. 7 (d-f). In these matrixes, 1–4 represent pure EVOO, EVOO&PO, EVOO&CO and EVOO&SO blends, respectively. At medium and low levels of adulteration, misclassification rates increased, which may be related to the high spectral similarity among different oil products. However, AlexNet still demonstrated the highest discrimination ability. These results confirm AlexNet’s strength in handling spectral data and its capacity to deliver more reliable identification in challenging EVOO adulteration tasks.

Figure 8 presents the average ROC curves and corresponding AUC scores obtained using three classifiers for detecting different types of adulterated EVOO. The AUC was 0.93–0.95 in group I, 0.89–0.92 in group II, and 0.82–0.88 in group III. The models perform strongly at high adulteration levels, yet their accuracy falls as the adulterant concentration drops. All three models showed similar overall performance in identifying olive oil with different levels of adulteration. AlexNet model exhibited the highest accuracy, attaining rates of 0.95, 0.92, and 0.88 at high, medium, and low adulteration concentrations, respectively.

Figure 9 shows the average accuracy for discriminating pure EVOO from adulterated EVOO and for identifying the specific vegetable oil used for adulteration. The analysis results indicate that CNN and LSTM models demonstrate relatively low recognition accuracy. Specifically, the CNN and LSTM models achieved accuracy rates of 82% and 86% in identifying low adulteration, compared to an accuracy rate of 88% for the AlexNet model. This advantage primarily stems from AlexNet's deeper convolutional and max-pooling layers, which are capable of extracting richer and more high-level features. Furthermore, the multi-output architecture allows the model to perform simultaneous qualitative identification of different oil types. As a result, AlexNet exhibited significantly better classification performance than the other two models.

Table 2 shows the sensitivity (Sn) and specificity (Sp) of the three models in identifying adulterated EVOO. The Sn and Sp of AlexNet in identifying adulterated EVOO are minimal differences, with extremely low risks of misclassification and missed classification. This indicates the strong ability of the model to distinguish between different levels of adulteration. The LSTM model followed closely behind, while the Sn and Sp of the traditional CNN model were lower than both models. AlexNet and LSTM outperform the conventional CNN in both generalization and noise robustness, which is attributed to their deeper convolutional hierarchies and gated temporal memory.

Table 2
Prediction indicators of each deep learning model

Model	Level	EVOO		EVOO&PO		EVOO&CO		EVOO&SO	
		[d]Sn(%)	[e]Sp(%)	Sn(%)	Sp(%)	Sn(%)	Sp(%)	Sn(%)	Sp(%)
CNN	pure	100	100	——	——	——	——	——	——
	HA	——	——	93.50	92.14	92.89	92.07	93.61	92.24
	MA	——	——	89.42	89.14	89.46	89.12	89.62	89.54
	LA	——	——	82.24	81.46	82.55	81.37	82.50	81.47
LSTM	pure	100	100	——	——	——	——	——	——
	HA	——	——	94.67	93.67	94.86	94.67	94.86	94.86
	MA	——	——	90.44	89.61	90.11	89.56	90.78	89.71
	LA	——	——	86.54	85.89	86.46	85.78	86.32	85.88
AlexNet	pure	100	100	——	——	——	——	——	——
	HA	——	——	95.62	95.24	95.46	95.26	95.32	95.11
	MA	——	——	92.26	91.78	92.44	91.98	92.12	91.89
	LA	——	——	88.67	87.85	88.55	88.02	88.21	87.69

[d]Sn: Sensitivity [e]Sp: Specificity

3.4. Quantitative analysis and evaluation of regression models

In this study, a spectral dataset of blended EVOO was selected, covering adulteration ratios from 5% to 50%, as illustrated in Table 1. The accuracy, stability, and applicability of the three regression models were systematically evaluated across the full range of adulteration ratios. The experimental results for EVOO-PO are displayed in Fig. 10. Predictions from both LSTM and AlexNet regressions are closely aligned with the true values and perform excellently across most scenes. The R^2 value of LSTM is 0.9811, and the RMSE value is 0.2954%; the R^2 value of AlexNet is 0.9930, and the RMSE value is 0.1258%. As illustrated in the figure, the majority of prediction values exhibit a high degree of proximity to the actual values. This confirms the accurate target-prediction ability of both AlexNet and LSTM. In comparison, the CNN model's prediction values demonstrate greater dispersion, with an R^2 value of 0.9615 and an RMSE value of 3.7255%. This phenomenon can be attributed to the spectral characteristics of EVOO-PO with low adulteration levels exhibiting a high degree of similarity.

Table 3 summarizes the regression performance of the three models on each adulterated EVOO set (100 replicate runs per sample to ensure statistical reliability). The results indicate that the AlexNet and LSTM models have significant advantages in quantitatively predicting the adulteration in olive oil. This can be attributed to the complex convolutional layers and max pooling layers of the AlexNet, which effectively extract data features. Due to the gated temporal processing of LSTM, which enhances long-term and short-term memory capabilities to mine information features from the data. Consequently, these two models demonstrate strong suitability for quantitative analysis of complex spatiotemporal sequences in adulterated olive oil.

Table 3
Results of regression analysis of sample concentration

Sample	CNN		LSTM		AlexNet	
	R^2	RMSE(%)	R^2	RMSE(%)	R^2	RMSE(%)
EVOO&PO	0.9615	3.7255	0.9811	0.2954	0.9930	0.1258
EVOO&CO	0.9621	3.7654	0.9815	0.2855	0.9931	0.1257
EVOO&SO	0.9581	3.8246	0.9804	0.4246	0.9935	0.1259

4. Conclusion

In this study, LIF technology combined with CNN, LSTM, and an improved AlexNet algorithm was systematically evaluated for its effectiveness in detecting adulteration in EVOO. Three models achieved 100% classification accuracy in distinguishing pure EVOO from adulterated samples. The AlexNet model achieved a recognition accuracy of 88%, even at low adulteration rates (< 5%), outperforming both CNN

and LSTM models and demonstrating high sensitivity in analyzing complex adulterated samples. Quantitative analysis revealed that the AlexNet and LSTM models were superior to the CNN model, with significantly higher R^2 and lower RMSE values (AlexNet: R^2: 0.9930, RMSE: 0.1258%; LSTM: R^2: 0.9811, RMSE: 0.2954%; CNN: R^2: 0.9615, RMSE: 3.7255%). This result positions them as more suitable tools for the precise prediction of olive oil adulteration concentration. The LIF-deep learning architecture proposed in this study enables qualitative identification of olive oil adulteration and quantitative prediction of adulteration concentration. This integrated approach serves as a novel methodology for olive oil analysis and paves the way for authenticating other complex food products.

Declarations

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This work was supported by Foundation of Shandong Province ZR2022QD113, Key R&D Program of Shandong Province 2024TSGC0704, Shandong Provincial Key Research and Development Program 2024CXGC010804. Project of Transportation Department of Shandong Province 2022B103.

Author Contribution

Li Le: Writing – original draft, Formal analysis, Software. Sun Lanjun: Writing – review & editing, Writing – original draft, Conceptualization. Meng Xiongfei: Investigation, Conceptualization. Liu Zhijian: Data curation. Huang Shuhan: Validation.

Data availability

Data will be made available on request.

References

1. Alharbi, H., Kahfi, J., Dutta, A., Jaremko, M., & Emwas, A. H. (2024). The detection of adulteration of olive oil with various vegetable oils–A case study using high-resolution 700 MHz NMR spectroscopy coupled with multivariate data analysis. *Food Control, 166,* 110679. https://doi.org/10.1016/j.foodcont.2024.110679

2. Arendse, E., Nieuwoudt, H., Magwaza, L.S. *et al.* Recent Advancements on Vibrational Spectroscopic Techniques for the Detection of Authenticity and Adulteration in Horticultural Products with a Specific Focus on Oils, Juices and Powders. *Food Bioprocess Technol 14*, 1–22 (2021). https://doi.org/10.1007/s11947-020-02505-x
3. Bavali, A., Rahmatpanahi, A., & Chegini, R. M. (2025). Quantitative analysis of the olive oil adulteration based on the assessment of multi-angle LIF spectral data using learning algorithms. *Journal of Food Composition and Analysis, 144,* 107723. https://doi.org/10.1016/j.jfca.2025.107723
4. Bongiorno, D., Di Stefano, V., Indelicato, S., Avellone, G., & Ceraulo, L. (2023). Bio-phenols determination in olive oils: Recent mass spectrometry approaches. *Mass Spectrometry Reviews, 42(4),* 1462–1502. https://doi.org/10.1002/mas.21744
5. Chen, Y., Zhang, J., Wang, J., Wang, C., & Zhu, Q. (2024). Analysis of small three-dimensional fluorescence spectroscopy dataset using migration learning an example of phenol concentration prediction in wastewater. *Science of The Total Environment, 954,* 176301. https://doi.org/10.1016/j.scitotenv.2024.176301
6. Fan, C., Yuan, X., & Zhang, Y. -J. (2019). CNN-Based Signal Detection for Banded Linear Systems. *IEEE Transactions on Wireless Communications, 18(9),* 4394–4407. https://doi.org/10.1109/TWC.2019.2924424
7. Gandhi, V., Roberts, P. J., Stoesser, T., Wright, H., & Kim, J. H. (2011). UV reactor flow visualization and mixing quantification using three-dimensional laser-induced fluorescence. *Water Research, 45(13),* 3855–3862. https://doi.org/10.1016/j.watres.2011.04.041
8. Hashempour-baltork, F., Zade, S. V., Mazaheri, Y., Alizadeh, A. M., Rastegar, H., Abdian, Z., & Damirchi, S. A. (2024). Recent methods in detection of olive oil adulteration: State-of-the-Art. *Journal of Agriculture and Food Research, 16,* 101123. https://doi.org/10.1016/j.jafr.2024.101123
9. Hooshyari, M., & Casale, M. (2021). Identification of Base Stock in Engine Oils by Near Infrared and Fluorescence Spectroscopies Coupled with Chemometrics. *Surveys in Geophysics, 42,* 451–465. https://doi.org/10.1007/s10712-020-09627-z
10. Izquierdo, M., Lastra-Mejías, M., González-Flores, E., Cancilla, J. C., Aroca-Santos, R., & Torrecilla, J. S. (2020). Deep thermal imaging to compute the adulteration state of extra virgin olive oil. *Computers and Electronics in Agriculture, 171,* 105290. https://doi.org/10.1016/j.compag.2020.105290
11. Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2017). ImageNet classification with deep convolutional neural networks. *Communications of the ACM, 60(6),* 84–90. https://doi.org/10.1145/3065386
12. Leng, T., Wang, Y., Wang, Z., Hu, X., Yuan, T., Yu, Q., Xie, J., & Chen, Y. (2025). Rapid classification of Camellia seed varieties and non-destructive high-throughput quantitative analysis of fatty acids based on non-targeted fingerprint spectroscopy combined with chemometrics. *Food Chemistry, 474,* 143181. https://doi.org/10.1016/j.foodchem.2025.143181.1

13. Liang, S., Chen, G., Ma, C. *et al.* Qualitative Identification and Adulteration Quantification of Extra Virgin Olive Oil Based on Raman Spectroscopy Combined with Multi-task Deep Learning Model. *Food Anal. Methods 18*, 385–397 (2025). https://doi.org/10.1007/s12161-024-02728-0
14. Li, D., Gao, H., Yang, K., Zhou, F., & Shi, X. (2023). Abnormal identification of oil monitoring based on LSTM and SVDD. *Wear, 526,* 204793. https://doi.org/10.1016/j.wear.2023.204793
15. Li, T., Liang, Z., Dreizler, A., & Böhm, B. (2023). Accurate determination of homogeneous ignition of single solid fuel particles enabled by machine learning. *Fuel, 338,* 127171. https://doi.org/10.1016/j.fuel.2022.127171
16. Lin, X. W., Li, F. L., Wang, S., et al. (2023). A Novel Method Based on Multi-Molecular Infrared (MM-IR) AlexNet for Rapid Detection of Trace Harmful Substances in Flour. *Food and Bioprocess Technology, 16,* 667–676. https://doi.org/10.1007/s11947-022-02964-4
17. Liu, Q., Dong, P., Fengou, L. C., Nychas, G. J., Fowler, S. M., Mao, Y., & Zhang, Y. (2023). Preliminary investigation into the prediction of indicators of beef spoilage using Raman and Fourier transform infrared spectroscopy. *Meat Science, 200,* 109168. https://doi.org/10.1016/j.meatsci.2023.109168
18. Loh, A., Ha, S. Y., Kim, D., Lee, J., Baek, K., & Yim, U. H. (2021). Development of a portable oil type classifier using laser-induced fluorescence spectrometer coupled with chemometrics. *Journal of Hazardous Materials, 416,* 125723. https://doi.org/10.1016/j.jhazmat.2021.125723
19. Malavi, D., Raes, K., & Van Haute, S. (2024). Integrating near-infrared hyperspectral imaging with machine learning and feature selection: Detecting adulteration of extra-virgin olive oil with lower-grade olive oils and hazelnut oil. *Current Research in Food Science, 9*, 100913, https://doi.org/10.1016/j.crfs.2024.100913
20. Mendes, T.O., da Rocha, R.A., Porto, B.L.S. et al. Quantification of Extra-virgin Olive Oil Adulteration with Soybean Oil: a Comparative Study of NIR, MIR, and Raman Spectroscopy Associated with Chemometric Approaches. *Food Anal. Methods 8,* 2339–2346 (2015). https://doi.org/10.1007/s12161-015-0121-y
21. Moraes, I. A., Kaufmann, K. C., Junior, S. B., Neves, M. G., Villa, J. E., & Barbin, D. F. (2025). Authentication of coriander oil and adulterant identification using electronic nose and spectroscopic techniques. *Food Chemistry, 483,* 144196. https://doi.org/10.1016/j.foodchem.2025.144196
22. Rahman, M. F. A., Elhawary, E., Hafez, A. M., Capanoglu, E., Fang, Y., & Farag, M. A. (2024). How does olive seed chemistry, health benefits and action mechanisms compare to its fruit oil? A comprehensive review for valorization purposes and maximizing its health benefits. *Food Bioscience*, *59*, 104017. https://doi.org/10.1016/j.fbio.2024.104017
23. Rohman, A., & Man, Y. B. C. (2011). Application of Fourier Transform Infrared Spectroscopy for Authentication of Functional Food Oils. *Applied Spectroscopy Reviews, 47(1),* 1–13. https://doi.org/10.1080/05704928.2011.619020
24. Saitta, M., Lo Curto, S., Salvo, F., Di Bella, G., & Dugo, G. (2002). Gas chromatographic–tandem mass spectrometric identification of phenolic compounds in Sicilian olive oils. *Analytica Chimica Acta, 466(2),* 335–344. https://doi.org/10.1016/S0003-2670(02)00572-X

25. Simard, J. -R., Roy, G., Mathieu, P., Larochelle, V., McFee, J., & Ho, J. (2004). Standoff sensing of bioaerosols using intensified range-gated spectral analysis of laser-induced fluorescence. *IEEE Transactions on Geoscience and Remote Sensing, 42(4),* 865–874. https://doi.org/10.1109/TGRS.2003.823285
26. Song, W., Song, Z., Vincent, J., Wang, H., & Wang, Z. (2020). Quantification of extra virgin olive oil adulteration using smartphone videos. *Talanta, 216,* 120920. https://doi.org/10.1016/j.talanta.2020.120920
27. Stavrakakis, G., Philippidis, A. & Velegrakis, M. Application of Optical Spectroscopic Techniques and Multivariate Statistical Analysis as a Method of Determining the Percentage and Type of Adulteration of Extra Virgin Olive Oil. *Food Anal. Methods 15*, 285–293 (2022). https://doi.org/10.1007/s12161-021-02055-8
28. Sun, Q., Liu, W., Wen, C., & Cai, L. (2024). Estimation of Marine Oil Spill Surface Characteristics Based on Laser-Induced Fluorescence With IA-DBSCAN. *IEEE Transactions on Instrumentation and Measurement, 73,* 1–14. https://doi.org/10.1109/TIM.2024.3396852
29. Tadini, A. M., Xavier, A. A., Milori, D. M., Oliveira, P. P., Pezzopane, J. R., Bernardi, A. C., & Martin-Neto, L. (2021). Evaluation of soil organic matter from integrated production systems using laser-induced fluorescence spectroscopy. *Soil and Tillage Research, 211,* 105001. https://doi.org/10.1016/j.still.2021.105001
30. Temiz, H. T., Velioglu, S. D., Guner, K. G., & Velioglu, H. M. (2021). The use of Raman spectroscopy and chemometrics for the discrimination of lab-produced, commercial, and adulterated cold-pressed oils. *LWT,146,* 111479. https://doi.org/10.1016/j.lwt.2021.111479
31. Torreblanca-Zanca, A., Aroca-Santos, R., Lastra-Mejías, M., Izquierdo, M., Cancilla, J. C., & Torrecilla, J. S. (2019). Laser diode induced excitation of PDO extra virgin olive oils for cognitive authentication and fraud detection. *Sensors and Actuators B: Chemical, 280,* 1–9. https://doi.org/10.1016/j.snb.2018.10.014
32. Uncu, O., Ozen, B., & Tokatli, F. (2019). Use of FTIR and UV–visible spectroscopy in determination of chemical characteristics of olive oils. *Talanta, 201,* 65–73. https://doi.org/10.1016/j.talanta.2019.03.116
33. Venturini, F., Sperti, M., Michelucci, U., Gucciardi, A., Martos, V. M., & Deriu, M. A. (2023). Extraction of physicochemical properties from the fluorescence spectrum with 1D convolutional neural networks: Application to olive oil. *Journal of Food Engineering, 336,* 111198. https://doi.org/10.1016/j.jfoodeng.2022.111198
34. Yang, J., Chang, B., Zhang, Y., et al. (2022). CNN coal and rock recognition method based on hyperspectral data. *International Journal of Coal Science and Technology, 9,* 63. https://doi.org/10.1007/s40789-022-00516-x
35. Yang, X., Pei, J., He, X., Wang, Y., Wang, L., Shen, F., Li, P., & Fang, Y. (2024). A novel method for determination of peroxide value and acid value of extra-virgin olive oil based on fluorescence

internal filtering effect correction. *Food Chemistry, 441,* 138342. https://doi.org/10.1016/j.foodchem.2023.138342

36. Yan, P., Li, G., Wang, W., Zhao, Y., Wang, J., & Wen, Z. (2024). Qualitative and quantitative detection of microplastics in soil based on LIF technology combined with OOA-ELM/SPA-PLS. *Microchemical Journal, 201,* 110632. https://doi.org/10.1016/j.microc.2024.110632
37. Zhang, Y., Li, T., Chen, H., Chen, S., Guo, P., & Li, Y. (2019). Excitation wavelength analysis of a laser-induced fluorescence technique for quantification of extra virgin olive oil adulteration. *Applied Optics, 58,* 4484–4491. https://doi.org/10.1364/AO.58.004484
38. Zhu, J., Su, W., Yin, C., Yue, J., & Wu, H. (2025). Identification for the Raman spectra of edible oil mixture with convolutional neural network. *Optics & Laser Technology,192(Part C),*113783. https://doi.org/10.1016/j.optlastec.2025.113783

Figures

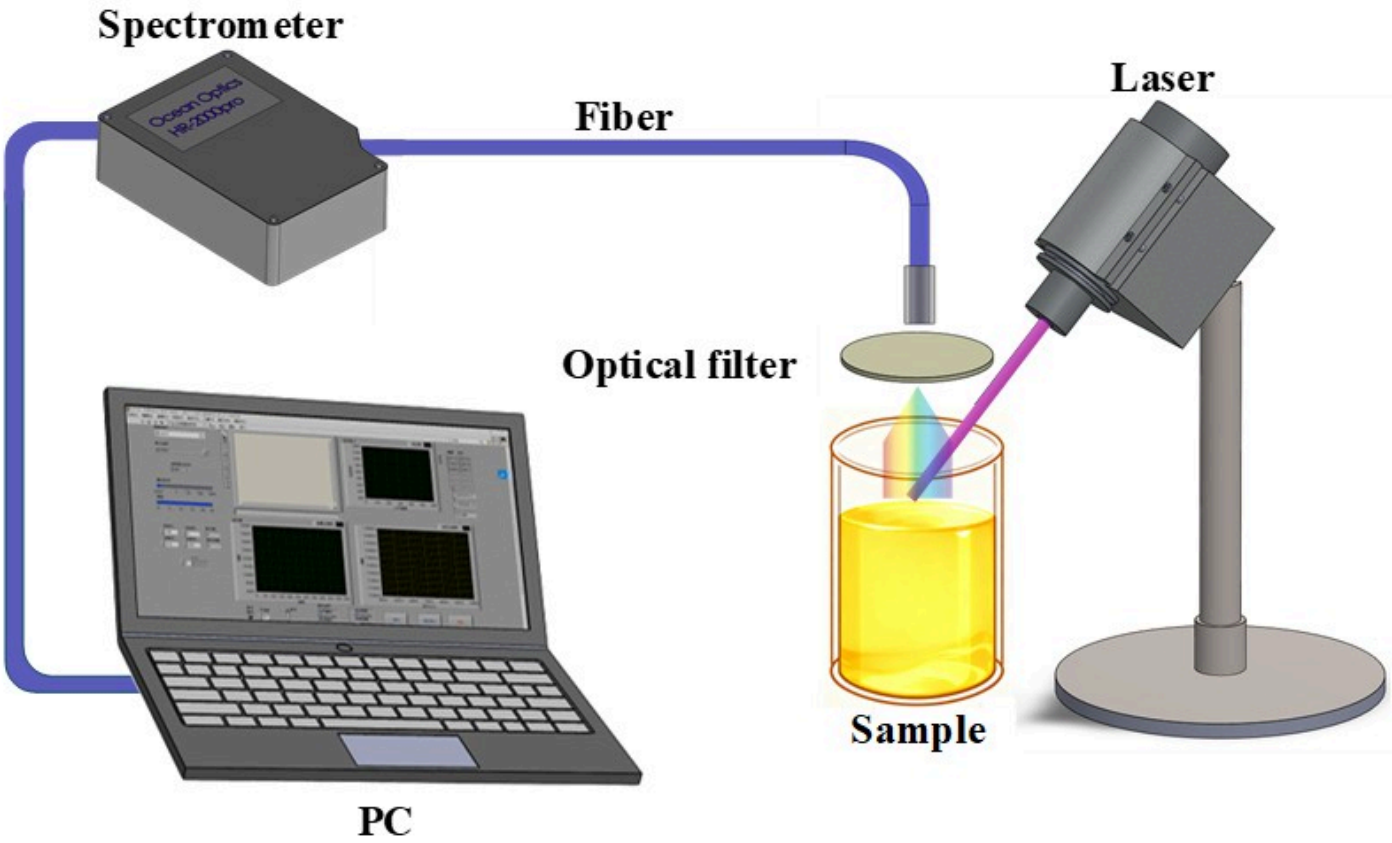


Figure 1

Schematic diagram of the LIF experimental setup

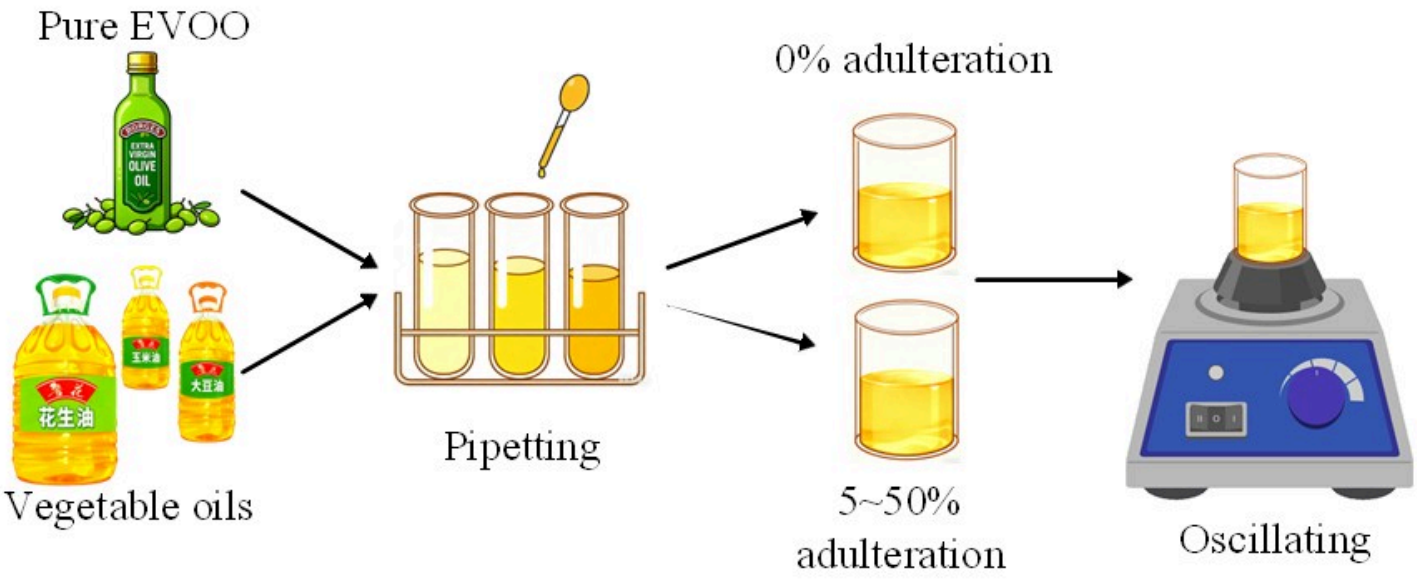


Figure 2

Experimental sample preparation process

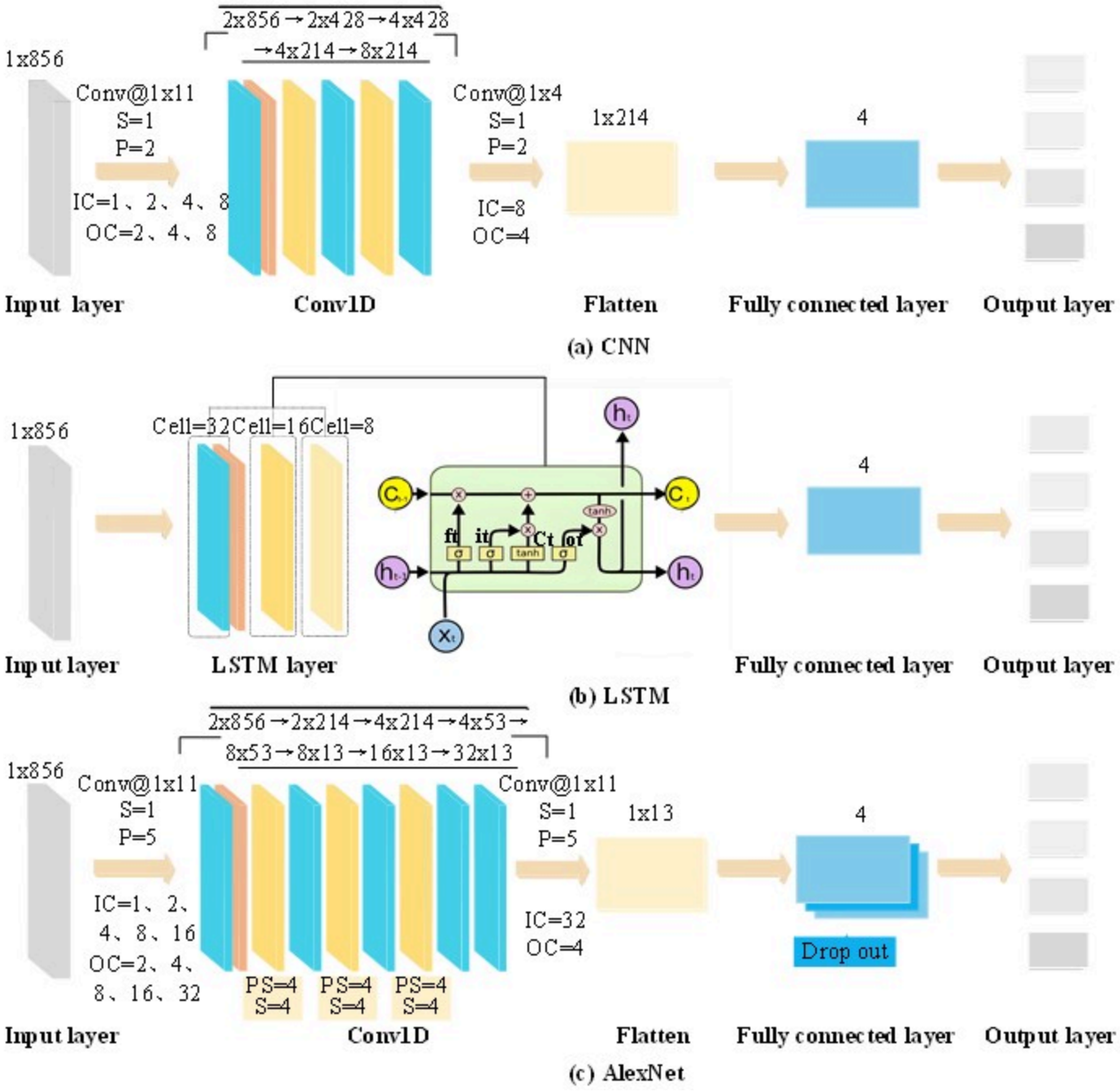


Figure 3

Structural diagrams of three deep network models, where orange represents BN layers. In Fig.(a), blue represents Conv layers and yellow represents pooling layers; in Fig.(b), blue, yellow, and light yellow represent three LSTM layers, each containing gate units for the forget gate, input gate, and output gate; in Fig.(c), blue represents Conv layers and yellow represents max pooling layers

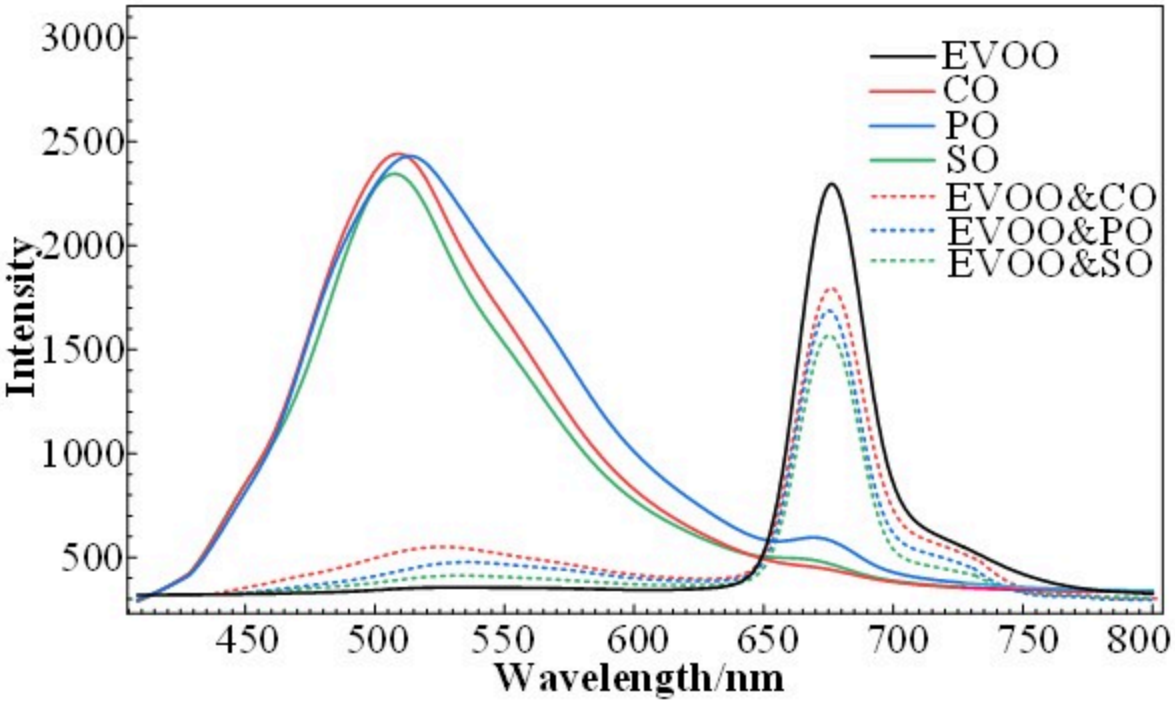


Figure 4

Fluorescence spectrum of oil sample

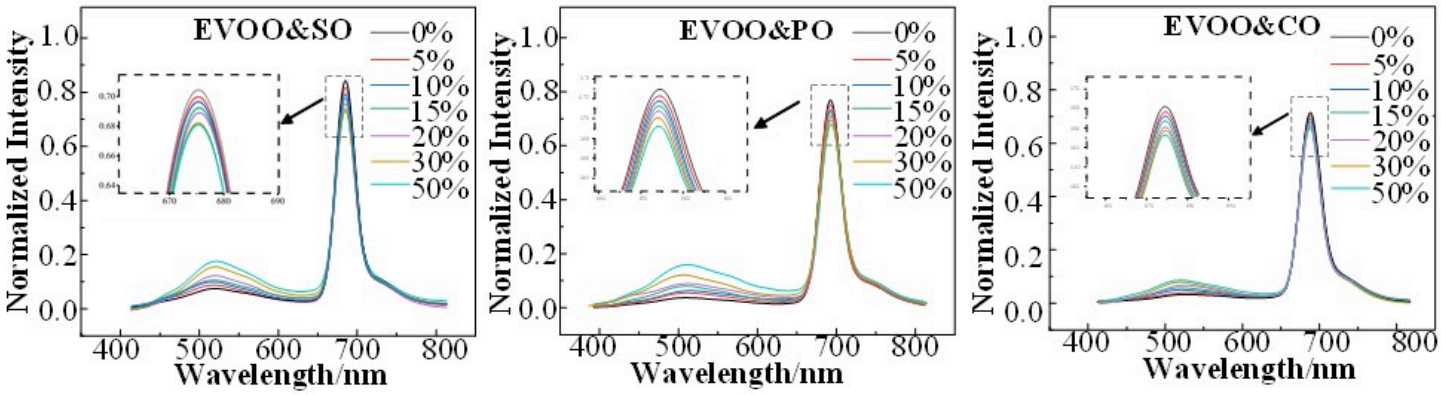


Figure 5

Fluorescence spectra of samples with different adulteration ratios

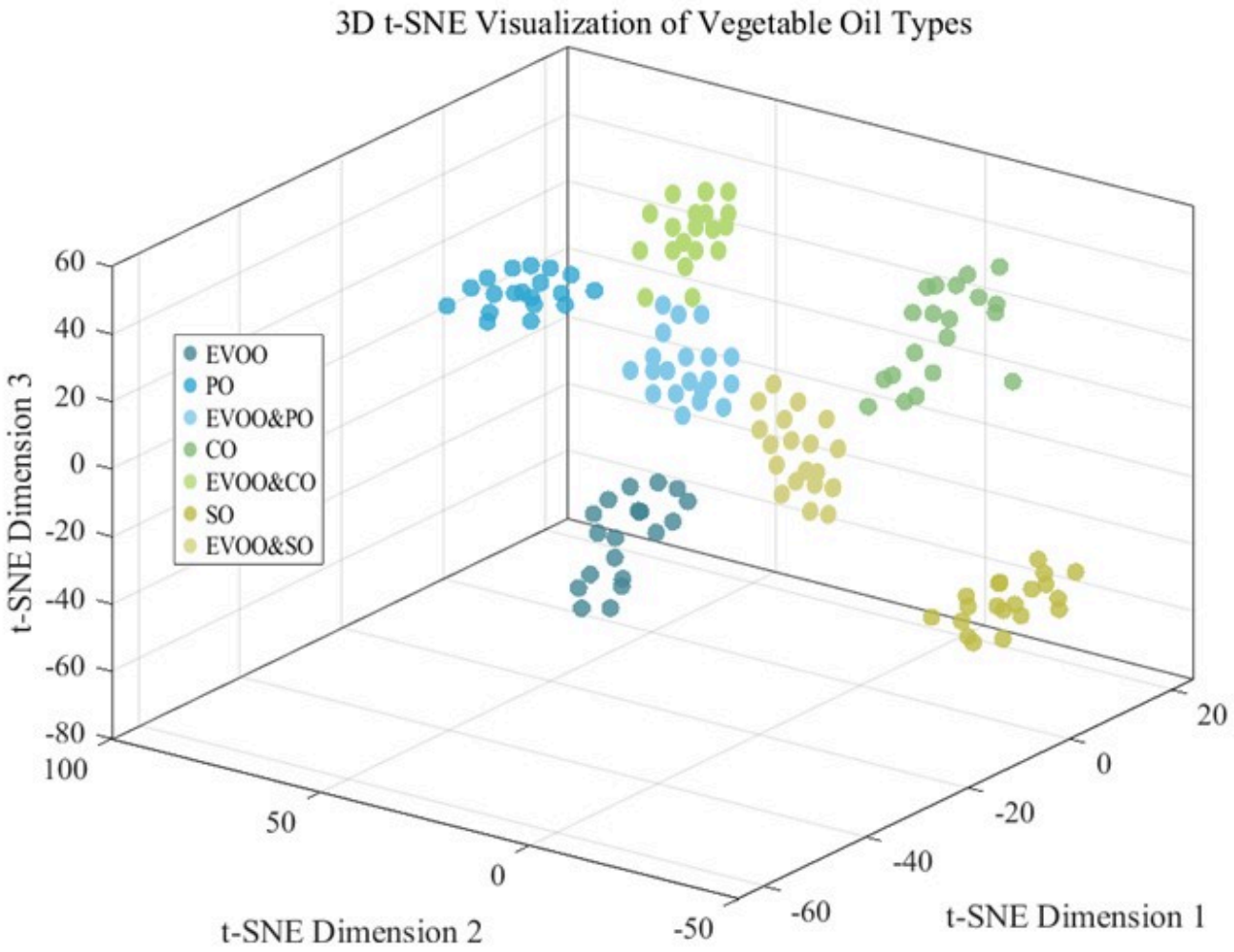


Figure 6

Three-dimensional t-SNE visualization of the seven vegetable oils

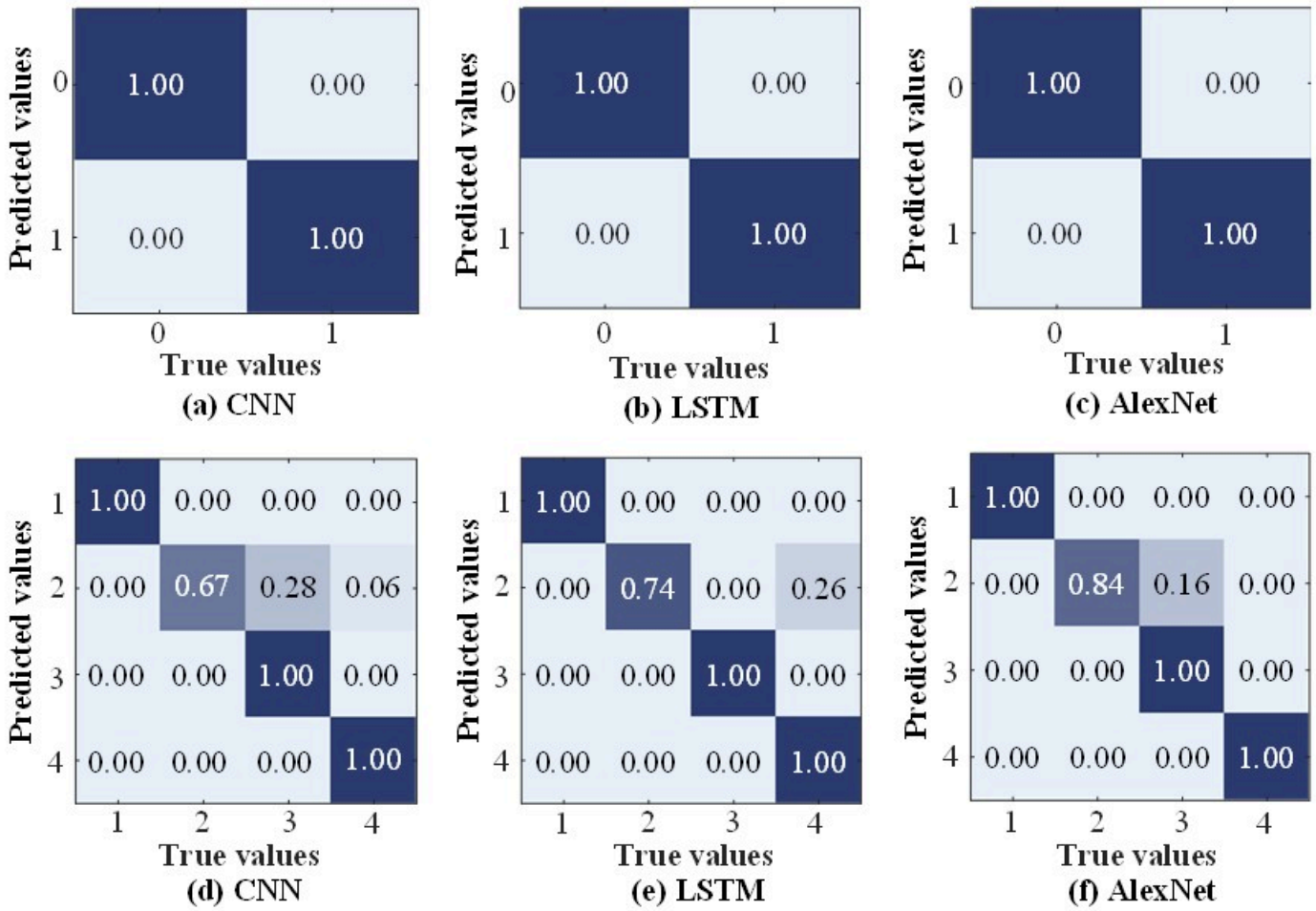


Figure 7

Confusion matrix diagram for EVOO classification

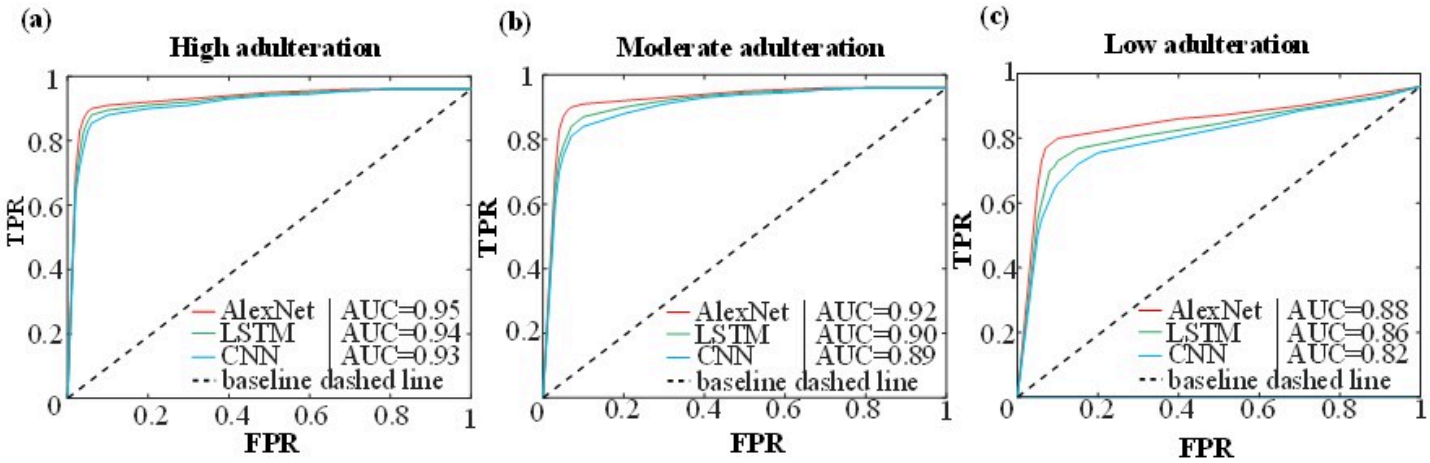


Figure 8

ROC and AUC for detecting EVOO adulteration at different levels

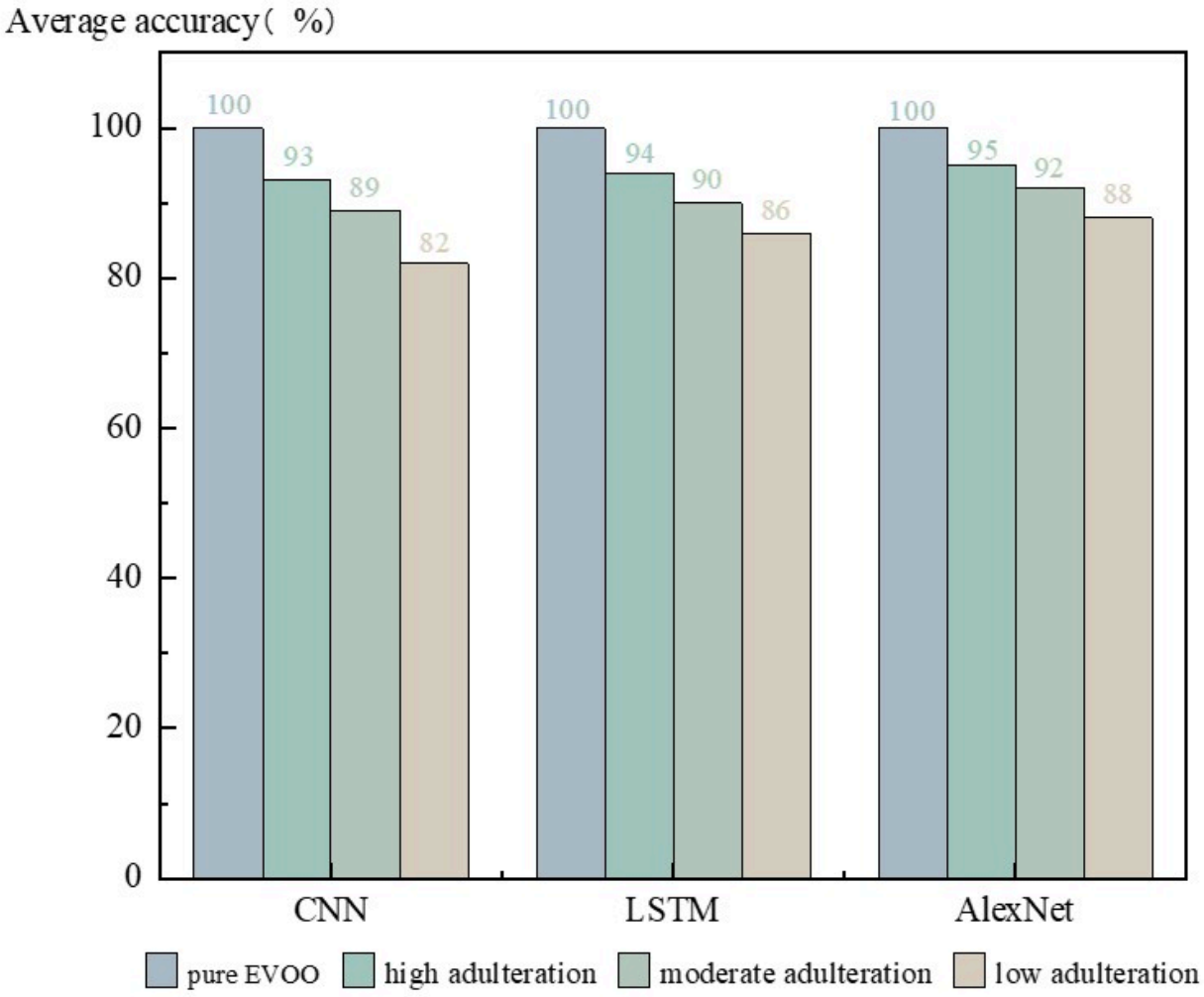


Figure 9

Average accuracy of classification models

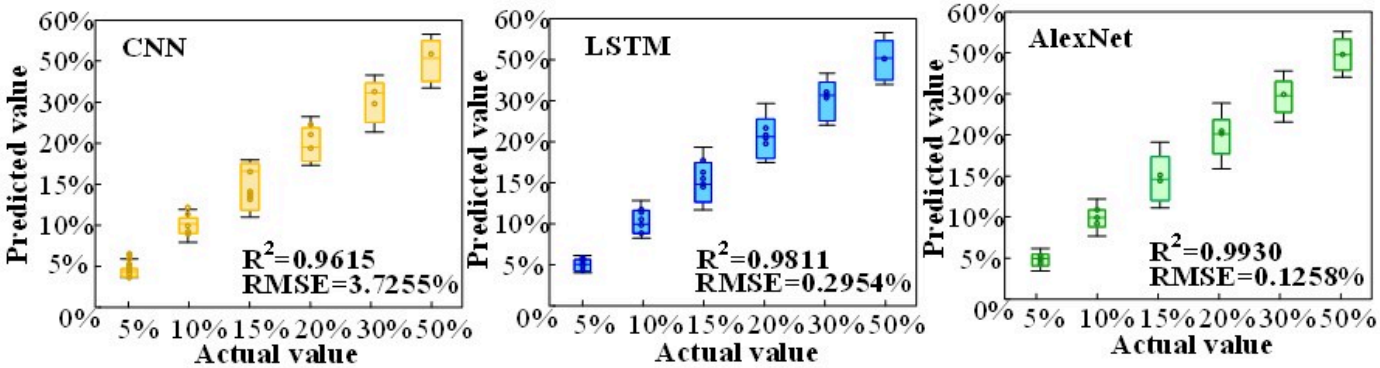


Figure 10

Model regression analysis results